
RESEARCH INTERESTS

My interests lie in the application of physics-informed machine learning, non-linear control & optimization, and dynamics modelling in fields such as the control of mobile manipulation, space robotics and autonomous systems. My research seeks to combine the flexibility of machine learning methods with a grounding in engineering, mathematics and physics to develop solutions to complex real-world problems in the manufacturing, agriculture and space sectors. This has led directed my research path to combining control, metamaterials and robotics for autonomous solutions.

CURRENT ROLE(S)

- **Post-Doctoral Research Associate** Sheffield, UK
School of Electrical & Electronic Engineering, University of Sheffield *Oct 2024 - Present*
 - **Projects:** Working as a research associate in systems & control on the STFC Space Systems grant, where my research is on multi-satellite data analysis and the attitude control & flight dynamics of autonomous spacecraft
 - **Previously:** Prior work in the School of Mechanical, Aerospace & Civil Engineering working on the EPSRC COATIN and STAMAN projects and in the EPSRC RESCu-M2 hub
 - **Research:** Work includes: spacecraft autonomy, in-orbit decision making, nonlinear dynamics estimation and prediction, control theory, metamaterial design and control
 - **Challenge Area Co-Lead, Space & Aviation:** Co-lead of the Space & Aviation Challenge Area for the UK Metamaterials Network
 - **Leadership:** Deputy Director of the RAMS Lab. Co-lead of the Applied & Theoretical Metamaterials Lab (ATM Lab)
- **Teaching Associate** Sheffield, UK
School of Electrical & Electronic Engineering, University of Sheffield *Aug 2025 - Present*
 - **Lecturing:** Module leader for teaching undergraduate and graduate students in robotics and control.
 - **Supervision:** Supervising undergraduate, graduate and postgraduate students in their research.
 - **FHEA:** Awarded Fellow of the Higher Education Academy (FHEA) for teaching and supervision.

GRANTS

- **Henry Royce Institute:** Industrial Collaboration Program Grant (*co-investigator*) - £88,285 *Mar 2026*
- **Airbus UK:** Robotics Research Scholarship (*co-investigator*) - £16,000 *Sept 2020, Aug 2021*
- **EPSRC:** ICASE Research Scholarship - £6,000 *Aug 2021*
- **RS Online:** Grassroots Student Project Grant (*principal investigator*) - £1,000 *Mar 2021*

EDUCATION

- **Doctorate of Philosophy (Ph.D.), Systems & Control** Sheffield, UK
School of Mechanical, Aerospace & Civil Engineering, University of Sheffield *Aug 2021 - Aug 2025*
 - **Research:** Undertook a PhD in the School of Mechanical, Aerospace & Civil Engineering, supervised by Prof. Ashutosh Tiwari and Dr. Simon Pope. Research is centred around learning-based methods for decision-making, planning and control for autonomous systems.
 - **Thesis Title:** *Learning-Based Methods for Decision-Making, Planning & Control* working in partnership with Airbus UK.
 - **Assessors::** Dr. Jonathan Aitken (internal) & Prof. Jeremy Watson (external, UCL)
- **Masters of Engineering (M.Eng.), Aerospace Engineering w/ Year in Industry** Sheffield, UK
School of Mechanical, Aerospace & Civil Engineering, University of Sheffield *Sept 2016 - Jul 2021*
 - **Studies:** Graduated with 1st Class (4.0 GPA) degree, studies in avionics, robotics & control theory.
 - **Graduate Thesis:** *Metrology-Assisted Assembly Process to Improve Landing Gear Installation Accuracy.*

PUBLICATIONS

- **CONFERENCE:** Maxwell Bird, **Ethan Canzini**, Simon Pope, “Onboard Relative State Estimation for Proximal Bodies in Orbit Using Recurrent Neural Networks”, *Under review at ESA SPAICE 2026*.
- **JOURNAL:** **Ethan Canzini**, Simon Pope, Ashutosh Tiwari, “Geometric Visual Servo via Optimal Transport”, June 2025. *Under Review at Control Engineering Practice*.
- **JOURNAL:** **Ethan Canzini**, David Miller, Divya Tiwari, Windo Hutabarat, Allan Matthews, Ashutosh Tiwari, “Analysis of Sensing Modalities for Electrode-Induction Gas Atomization of Metal Powders”, in *Surface & Coating Technologies*.
- **JOURNAL:** Néstor Sanchez-Arriaga, **Ethan Canzini (Joint first author)**, Nathan Plumb, Michael Farnsworth, Adrian Leyland, Ashutosh Tiwari, “Enhancing Robotic Wafer Inspection with Sensor Fusion and Learned Manifolds”, in *Nature Scientific Reports*.
- **CONFERENCE:** **Ethan Canzini**, Simon Pope, Ashutosh Tiwari, “Generating Continuous Paths On Learned Constraint Manifolds Using Policy Search”, *IEEE/RSJ International Conference on Intelligent Robots & Systems (IROS)*, Abu Dhabi, UAE, October 2024.
- **JOURNAL:** **Ethan Canzini**, Marc Auledas-Noguera, Simon Pope, Ashutosh Tiwari, “Decision Making For Multi-Robot Fixture Planning Using Multi Agent Reinforcement Learning”, in *IEEE Transactions on Automation Science & Engineering*.
- **CONFERENCE:** **Ethan Canzini**, Marc Auledas-Noguera, Dominique Chasteau and Ashutosh Tiwari, “A Novel Sensing Template Using Data Fusion for Large Volume Assembly”, *14th IFAC Workshop on Intelligent Manufacturing Systems*, Tel Aviv-Yafo (Israel), March 2022.

EMPLOYMENT HISTORY

Industrial Experience

- **Teaching Assistant** Sheffield, UK
University of Sheffield Sept 2021 - Jul 2024
 - **Role:** Working as a Graduate Teaching assistant (GTA) for a variety of modules within the faculty for undergraduate, graduate and postgraduate students.
 - **Modules:** Embedded systems, optimal control, hardware-in-the-loop control, aerospace automation, control theory & robotics.
- **Research Scientist** Broughton, UK
Airbus Operations Ltd. Sept 2021 - Jul 2024
 - **Role:** Research scientist working on developing autonomous robot solutions for manufacturing operations.
 - **Contribution:** Assisted on the development and deployment of robotic fixtures system for wing assembly.
 - **Dissemination:** Presented work package as part of the *Made Smarter Centre for Connected Factories*.
- **Automation & Robotics Engineer** Weymouth, UK
ASM Assembly Systems Ltd. Jul 2019 - Jul 2020
 - **Role:** Worked on developing and deploying robotic systems for semiconductor manufacturing.
 - **Autonomous AGV:** Developed a SLAM-guided AGV for the transportation of large parts and machines.
 - **ML Fault Detection System:** Maintained the SQL server used to store all machine test data and wrote the machine learning back-end to predict faults in manufacturing processes.
- **Instrument & Test Engineer** Durham, UK
CAV Ice Protection Systems Jul 2017 - Sept 2017
 - **Role:** Worked on building, testing and integrating aerospace instruments.
 - **“Iron bird” Design:** Managed the deployment of the “Iron bird” digital twin of aircraft control surfaces.
 - **Test Hardware Development:** Build the software and hardware for evaluating control surfaces and systems with integrated heating elements.

Internships

- **Research Intern** Didcot, UK
Growbotics Ltd. *Jul 2024 - Sept 2024*
 - **Role:** Research intern in the field of autonomous satellites & robotics for sustainable space exploration
 - **Collaboration:** Liaised with 3rd party companies and launch providers with producing space-suitable hardware and control interfaces
 - **Satellite State Estimation:** Developed low-cost sensor-based state estimators for full 6-DOF satellites using microcontrollers.
- **Visiting Researcher** Zürich, Switzerland
ETH Zürich *Jul 2022*
 - **Role:** Invited to be a visiting researcher at ETH Zurich as part of the Robotics Summer School.
 - **Topics:** Trajectory optimisation, multi-sensor fusion, SLAM, navigation, full-stack ROS and open source software for deployable robots.

TEACHING EXPERIENCE

Module Teaching (at UoS)

- **Hardware-in-the-Loop & Rapid Control Prototyping (ELE308):** 2025-Present
Module leader and lecturer for undergraduate control systems module focusing on deploying real-time controllers to hardware using LabVIEW and MATLAB. Developed the lecture & assessment content.
- **Rapid Control Prototyping (ELE438):** 2025-Present
Module leader and lecturer for graduate control systems module focusing on developing optimal control algorithms on embedded hardware. Developed the lecture & assessment content.
- **Industrial Training Programme, Avionics (ACS402):** 2021-2023
Module leader and lecturer for graduate avionics module in partnership with Airbus UK. Developed the lecture & assessment content and planned the Airbus interaction.
- **Real-Time Embedded Systems (ACS6127):** 2021-2024
Teaching Assistant and Module Leader for graduate control systems module for developing real-time embedded systems. Led labs and tutorial sessions, and delivered lecture content.

Supervision (at UoS)

- **Postgraduate:**
Nathan Espley-Plumb, *Bottom-up Connector-based Assembly for Flexible Assembly Systems*, Ph.D., Feb 2023 - Present. In partnership with Siemens Plc.
- **Graduate:**
Dhiraj Jandu, *Using Force-Torque Sensing to aid Peg-in-Hole Assembly*, M.Eng. thesis, 2026. In partnership with Airbus UK
Joaquin Barawed, *Accurate Feature Detection and Tracking for Relative Pose Estimation in Assembly*, M.Eng. thesis, 2025
Jake O'Neill, *Intelligent Inspection System for Adaptive Robotic Scanning*, M.Sc. thesis, 2025
Elena Manoli, *Towards Industry 5.0: Dynamic Robot Behaviour Manipulation Based on Human Proximity and Movement*, M.Sc. thesis, 2024
- **Undergraduate:**
William Santoso, *Geometric Control of Robot Manipulators*, B.Eng. dissertation, 2026
Maxwell Bird, *Relative Dynamics Estimation of Orbiting Spacecraft*, B.Eng. dissertation, 2025

ACHIEVEMENTS

Invited Talks

- **“Decision Making for Multi-Robot Fixture Planning Using Multi-Agent Reinforcement Learning”:** May 2025
Presented at the 2025 IEEE International Conference on Robotics & Automation (ICRA), Georgia Institute of Technology
- **“Geometry-Informed Systems for Robotic Manipulation: Methods in Planning & Control”:** May 2025
Presented at the meeting of the Dynamics Research Group, School of Mechanical, Aerospace & Civil Engineering, University of Sheffield

- “Scaling Robotic Capability In Industry Using Multi-Agent Systems - Applications in Agriculture, Construction & Manufacturing”:
Presented at the National Polytechnic Institute of Mexico, Ciudad de México, Mexico

June 2023

Professional Activities

- **HEA:** Fellow of the Higher Education Academy (FHEA) 2024 - Present
- **IEEE:** Member of the Institute of Electrical & Electronic Engineers (MIEEE) 2020 - Present